

what happens **behind the scenes** when you run each of the core Terraform commands: `terraform init`, `terraform plan`, and `terraform apply`.

1. `terraform init` – Initialization

This command prepares working directory for other Terraform commands. It's **always the first command** run in a new Terraform project.

Backend process:

1. Loads configuration files (`.tf` files)
2. Initializes the backend
3. Downloads the required provider plugins
4. Configures provider settings
5. Checks for required Terraform version

Output:

Creates a `.terraform/` directory and stores:

- Provider binaries
- Backend info
- Plugin cache
- Dependency lock file (`.terraform.lock.hcl`)

2. `terraform plan` – Execution Plan

Generates an execution plan showing what Terraform *will do* if you run `apply`.

Backend process:

1. Loads all configuration files
2. Reads the current state
3. Refreshes the state
4. Compares desired vs current
5. Generates the plan

3. terraform apply – Apply Changes

Applies the changes required to reach the desired state defined in your `.tf` files.

Backend process:

1. Re-runs `terraform plan` logic
2. Prompts for approval (if not using `-auto-approve`)
3. Sends requests to the provider
4. Waits for completion
5. Updates the Terraform state file

Summary of Backend Flow

Command

What Happens Internally

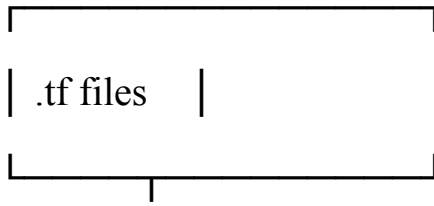
terraform init Downloads providers, configures backend, prepares dire

terraform plan Loads config, refreshes real state,
compares with desired state, shows changes

terraform apply Replans (if needed), makes
API calls to update infra, updates state file



Visual Summary



terraform init



- Downloads provider plugins
- Initializes backend



terraform plan



- Reads & refreshes current state
- Compares to .tf files
- Generates plan (diff)



terraform apply



- Executes plan (calls cloud APIs)
- Creates/updates/destroys resources
- Updates state file